

CS F212 DATABASE SYSTEMS Sem II 2024-25

LAB 4 EXERCISE (SUBQUERIES AND JOINS)

Total marks: 10 (1 mark per question). Weightage: 5%

Instructions: The solution can be found in multiple ways. Today's lab is based on SubQueries and Joins. Thus solutions using SET operations, though correct, will not be considered for evaluation.

Solve Q1 to Q5 using nested queries (subqueries) concept only. No marks will be given if solved using SET or JOINS concepts.

Q1. The annual Car Scale Model Fans' Meet is organised by offices of the company located in cities where at least one customer lives. Display, **in alphabetical order**, the cities where the meet will be held.

```
+-----+
| city   |
+-----+
| Boston |
| London |
| NYC    |
| Paris  |
| San Francisco |
+-----+
```

Q2. Display, **in alphabetical order**, the names of the products that have never been ordered (i.e., are absent from `orderdetails`) without the set operation.

```
+-----+
| productName |
+-----+
| 1985 Toyota Supra |
+-----+
```

Q3. Which customer(s) made the largest single payment (by `amount`)? The challenge is that you must answer this **without using the aggregate function**.

```
+-----+
| customerName |
+-----+
| Euro+ Shopping Channel |
+-----+
```

Q4. Citibank has called the company informing them that check no. CITI3434344 has bounced (see: payments). The customer who gave that check would like to visit one of the company's offices to sort out the issue. **In alphabetical order**, list the cities where the offices in that customer's country are located.

```
+-----+
| city   |
+-----+
| Boston |
| NYC    |
| San Francisco |
+-----+
```

Q5. List, in **ascending alphabetical order**, the names of all products that were ordered in 2005 by customers in the city Kobenhavn*.

**Danish spelling of Copenhagen*

```
+-----+
| productName |
+-----+
| 1948 Porsche Type 356 Roadster |
| 1993 Mazda RX-7 |
| 1995 Honda Civic1 |
+-----+
```

Solve the following Questions from Q6 to Q10 using the Joins concept only. No marks will be given if solved using SET operations or subqueries

- Q6 : Solve Q1
- Q7 : Solve Q2
- Q8 : Solve Q3
- Q9 : Solve Q4
- Q10 : Solve Q5